

Manjushree Aithal, PhD

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EDUCATION

PhD in Electrical and Computer Engineering | *Binghamton University, NY* | October 2022

Dissertation: "Mitigating Adversarial Attacks in Deep Learning Systems for Critical Applications"

Advisor: Dr. Xiaohua Li | GPA: 3.9/4.0

ME in Electronics & Telecommunication | *Pune University, India* | September 2015

Thesis: "Speech Enhancement Using PCA for Emotion Recognition"

BE in Electronics & Telecommunication | *Shivaji University, India* | May 2012

Magna Cum Laude

RESEARCH INTERESTS

Core Expertise: Computational research with proven experience in experiment design and hypothesis testing, data processing, A/B testing and performance benchmarking methodologies, collaborative research across interdisciplinary teams, scientific communication through peer-reviewed publications and conference presentations, and development of interactive tools and user interfaces for AI systems

Technical Skills: Advanced machine learning and deep learning implementation (PyTorch, TensorFlow, Keras), large language model fine-tuning using LoRA and PEFT frameworks, local agent optimization and on-device deployment, multi-agent framework development, model compression and optimization for resource-constrained deployment, computer vision and generative AI systems (GANs), adversarial robustness and AI system security, cloud infrastructure and scalable deployment pipelines (AWS, CI/CD), and predictive modeling with high-precision analytics

Research Focus: Applying established expertise in AI system development, optimization, and user-centered design to human-centered clinical AI applications, with active research in privacy-preserving clinical language models and multi-agent requirement-elicitation frameworks for clinical decision-making and healthcare informatics

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow – Clinical AI Framework Development

The Mitchell Lab, Department of Biomedical Informatics, University of Colorado Anschutz |
December 2025 – Present

- **PLACID - Privacy-Preserving Clinical Acronym Disambiguation:** Designed and evaluated a privacy-preserving cascaded pipeline enabling clinical language model deployment under strict healthcare data-privacy constraints
 - Architected a two-stage cascade routing general-purpose instruction-following models for clinical acronym detection to domain-specific biomedical models for context-relevant expansion
 - Benchmarked small-parameter (2B-10B) locally deployed models across detection and expansion tasks, achieving ~0.988 detection accuracy and raising expansion accuracy from ~0.655 to ~0.81 through the cascaded design
 - Demonstrated that on-device inference meets clinical accuracy requirements without external data transfer, addressing the privacy barrier that blocks cloud-based LLM integration in healthcare settings
- **LadderTeam - Dual-Agent Elicitation Framework:** Architected a dual-agent (Interviewer & Judge) framework automating manual, skill-dependent laddering interviews for requirement elicitation
 - Built a unified interview engine supporting ACV, 5-Whys, and JTBD probing methods over a shared seed and clarification phase
 - Implemented a reproducible five-step per-turn loop with state gating, drift detection, and abort logic, paired with a background Judge agent that scores each exchange without interrupting interview flow
 - Designed a controlled-simulation evaluation with scripted ground-truth transcripts isolating probe quality as the sole experimental variable, spanning 216 interviews across 2 personas, 4 models, 3 probing methods, and 3 UI scenarios
 - Achieved 99.1% chain convergence (214 of 216 runs), 81.0% ground-truth actionable match, and zero drift across all runs
- **PersonaDrift - Black-Box Persona Stability Measurement:** Designing a framework evaluating & measuring whether personality-calibrated language model personas hold their defined character across sustained multi-turn interaction, motivated by standardized-patient simulation for clinical training
 - Designed a three-channel drift metric - NEO-FFI personality trait score, behavioural deflection signature, and response quality i.e., measured entirely from behavioural output, requiring no internal activation access and therefore applicable to closed-API models
 - Implemented the behavioural channel as sentence-transformer cosine distance between each response and fixed persona-signature and neutral-baseline reference embeddings, providing a black-box analog to activation-space drift measurement
 - Built a purpose-designed evaluation judge scoring task-appropriateness and persona-consistency on sampled rounds, with the judge model held distinct from the persona and interviewer models to prevent self-evaluation bias

- Design 50 distinct agentic personas by mining a 421,000-post personality-typed corpus for per-type linguistic style and deriving trait targets through published MBTI-to-Big-Five correspondence mapping, confirmed by NEO-FFI calibration [work targeted for submission to AAMAS 2027 Main Technical Track]
- **MiniGali - Local Knowledge Agent Development (CU Innovations Fellowship):** Developing a fully local, privacy-preserving retrieval-augmented agent that captures institutional program knowledge for internal decision support, with privacy-first deployment for sensitive data

Staff Researcher – Mobile-Device AI Applications

Lenovo Research, IL | January 2025 – October 2025

- **Advanced Model Fine-Tuning:** Implementing Low-rank Adaptation (LoRA) techniques using PEFT framework for mobile feature summarization fine-tuning of Llama-3.2-3B-Instruct model, demonstrating expertise in adapting large language models for specialized applications
- **Mobile Deployment Optimization:** Investigating model compression strategies including post-training quantization and speculative decoding for mobile deployment across diverse hardware architectures (Mediatek & Qualcomm chipsets)
- **Scalable Model Pipeline Development:** Developing common cluster environment for model deployment pipeline with variations in model preparation and quantization techniques, showcasing skills in systematic approach to AI system deployment
- **Performance Trade-off Analysis:** Evaluating performance trade-offs across different hardware architectures, demonstrating analytical skills essential for optimizing AI systems for real-world constraints

AI Camera Researcher - Computer Vision Systems

Lenovo Research, IL | November 2022 – December 2024

- **Enterprise AI Feature Development:** Successfully delivered two mission-critical features (blink & gaze correction solution) for enterprise Gen-AI platform within strategic delivery timeframes, demonstrating project management and deadline-driven research capabilities
- **Advanced Model Architecture Design:** Engineered advanced generator architecture for enterprise Gen-AI solution, integrating lightweight Super Resolution module (8ms latency, +71MB GPU) while researching optimization techniques for mobile deployment directly transferable to developing efficient AI systems for resource-constrained environments
- **Scalable AI Infrastructure Development:** Engineered scalable in-house novel GAN model infrastructure for image enhancement with CI/CD integration, improving model training efficiency by 40% and implementing annotation automation that reduced manual QA overhead by 85%
- **Experiment Design and Pre-processing:** Performed exploratory architecture development for image fusion through iterative testing protocols, resulting in 35% improved information retrieval in high-dynamic-range scenarios
- **User-centred Performance Benchmarking:** Established standardized performance benchmarking methodology for ML model evaluation across device portfolio,

including comprehensive logging systems, A/B testing applications with intuitive UI/UX interface and performance analytics dashboards, reducing testing cycle time from 6 to 2 hours

- **Technical Documentation and Requirements Translation:** Formulated technical requirements documentation for AI solutions, translating business needs to executable specifications
- **Production-Ready AI Systems:** Implemented production-ready ML solutions using Python, TensorFlow, and AWS with focus on scalability, performance optimization, and cost management
- **Research Leadership and Innovation:** Successfully led and submitted 2 patents & 1 research paper for low-light image enhancement, demonstrating research excellence and innovation capabilities
- **Professional Recognition:** Recipient of company-wide SVP Individual Excellence Award for FY24/25 in recognition of outstanding contributions to image processing automation and enterprise Gen-AI feature development

RESEARCH EXPERIENCE

Graduate Research Fellow - AI Robustness and Security

Binghamton University, NY | August 2019 – October 2022

- **Novel Defense Algorithm Development:** Developed robust defense algorithms against state-of-the-art adversarial attacks, achieving 100% success rate in attack mitigation - demonstrating expertise in developing reliable AI systems for critical applications
- **Comprehensive Model Evaluation:** Analyzed and optimized defense methods by evaluating current and future complex problems in deep learning models, showcasing systematic approach to AI system reliability
- **Advanced Framework Implementation:** Implemented attack and defense methods using state-of-the-art adversarial toolboxes (Cleverhans, ART) in TensorFlow & PyTorch with GPU optimization
- **Systematic Validation Research:** Analyzed reliability of developed methods using adaptive attack approaches, demonstrating rigorous experimental design and validation methodologies
- **Generative Model Research:** Researched and developed pretrained VAE and DCGAN-generated approaches for solving query optimization problems in AI systems

Data Science Research Fellow - Predictive Analytics

Binghamton University, NY | March 2021 – May 2022

- **Predictive Model Development:** Developed data-driven artificial neural network models for predicting material behaviour on chip-set in vacuum environment over time with industrial applications
- **Comparative Machine Learning Analysis:** Systematically evaluated performance of ANN models against other ML approaches (SVR, logistic regression), demonstrating thorough experimental methodology
- **High-Accuracy Modelling:** Achieved 99.82% accuracy in predictive models for time-sensitive applications, showcasing expertise in developing precise, reliable AI systems
- **Efficiency Optimization:** Developed models reducing experimental validation time by 7 months, demonstrating understanding of practical deployment and workflow optimization

PUBLICATIONS & RESEARCH DISSEMINATION

Peer-Reviewed Publications

1. **Aithal, M.**, Kotz, A., & Mitchell, J. (2026). LadderTeam: Dual-Agent Laddering Elicitation Framework. *arXiv preprint arXiv:2608.17029*.
2. **Aithal, M.**, Kotz, A., & Mitchell, J. (2026). PLACID: Privacy-preserving Large language models for Acronym Clinical Inference and Disambiguation. *arXiv preprint arXiv:2603.23678*.
3. **Aithal, M.**, Vidal Mata, R.G., Kartha, M., Chen, G., Adhikarla, E., Kirsten, L.N., Fu, Z., Madhusudhana, N.A., Nasti, J.V. (2026). "LENVIZ: A High-Resolution Low-Exposure Night Vision Benchmark Dataset." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2531-2540.
4. Adhikarla, E., Zhang, K., VidalMata, R. G., **Aithal, M.**, Madhusudhana, N. A., Nicholson, J., ... & Davison, B. D. (2024, December). Unified-EGformer: exposure guided lightweight transformer for mixed-exposure image enhancement. In *International Conference on Pattern Recognition* (pp. 260-275). Cham: Springer Nature Switzerland.
5. **Aithal, M.**, Li, X. (2022). "Mitigating Black-Box Adversarial Attacks via Output Noise Perturbation." *IEEE Access Journal*, 10, 12395-12411.
6. **Aithal, M.**, Li, X. (2022). "Boundary Defence Against Black-box Adversarial Attacks." *26th International Conference on Pattern Recognition (ICPR)*, 2349-2356.
7. **Aithal, M.B.**, Sahare, S.L. (2015). "Emotion Detection from Distorted Speech Signal using PCA-Based Technique." *International Journal of Innovative Research in Advanced Engineering*, 2(9).
8. Gaikwad, P.R., **Aithal, M.B.**, Sahare, S.L. (2015). "Speech Enhancement Using PCA for Speech and Emotion Recognition." *Global Journal of Engineering, Design and Technology*, 4(3), 6-12.

Posters & Presentations

1. **Aithal, M.B.**, Kotz, A., Mitchell, J. (2026). "LadderTeam: Dual-Agent Laddering Elicitation Framework." *ACM AI Leadership Summit*, Atlanta, GA | Poster Presentation
2. **Aithal, M.B.**, Kotz, A., Mitchell, J. (2026). "PLACID: Privacy-preserving Large Language Models for Acronym Clinical Inference and Disambiguation." *AMIA Symposium* | Poster Presentation
3. **Aithal, M.**, Vidal Mata, R.G., Kartha, M., Chen, G., Adhikarla, E., Kirsten, L.N., Fu, Z., Madhusudhana, N.A., Nasti, J.V. (2026). "LENVIZ: A High-Resolution Low-Exposure Night Vision Benchmark Dataset." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson AZ | Poster Presentation

Under Review & Patent Applications

- **Patent Application:** "Machine Learning – based Prediction of Tin Whisker Formation and Penetration Time under Conformal Coatings" | Under Review
- **Patent Application:** "A Real-Time user Interactive AI Camera System in Mobile Devices" | Under Review
- **Patent Application:** "PLACID: Privacy-preserving Large language models for Acronym Clinical Inference and Disambiguation" | Under Review

INTERDISCIPLINARY COLLABORATION & INDUSTRY EXPERIENCE

Data Science Research Fellow

Digital DNA - Charge CCCV, NY | August 2018 – May 2022

- **Database Management Systems:** Designed and optimized novel database management system for collection and filtering of raw data from blockchain databases, demonstrating expertise in large-scale data handling and processing
- **Interdisciplinary Technology Integration:** Investigated and built digitizing blockchain systems and implemented deep learning for virtual designing of lithium-ion batteries, showcasing ability to work across multiple technical domains
- **Predictive Modeling Development:** Developed mathematical models for virtual battery design and analyzed performance using artificial neural networks, demonstrating strong analytical and modeling capabilities
- **Data Translation and Analysis:** Translated lithium-ion battery chemistry parameters into meaningful data for predictive modeling, showing ability to work with complex technical data and derive actionable insights
- **Multi-Platform Development:** Designed and implemented deep learning toolkits using R-programming and SaaS platforms for Digital DNA tool development
- **Technical Communication:** Worked as data analyst and web content developer for www.lithiumionbattery.org, demonstrating science communication skills

Hardware Engineering Research Intern

Universal Instruments Co, NY | January 2018 – June 2018

- **Precision System Development:** Developed high-speed wafer feeder systems using Xilinx platforms, implementing switching mechanisms between spindles for component placement with optimized speed and positioning control
 - **Embedded Systems Programming:** Developed working code for FIFO, power control, and spindle switching systems with comprehensive testing protocols
 - **Signal Processing and Optimization:** Designed digital filters for noise cancellation in high-speed systems, demonstrating signal processing expertise applicable to biomedical applications
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TECHNICAL SKILLS FOR AI RESEARCH

AI/ML Frameworks: PyTorch, TensorFlow, TensorFlow Lite, Keras, Pandas - *extensive experience in production deployment*

Programming Languages: Python, C, C++, MATLAB, R-programming, AWS - *full-stack development capabilities*

Specialized Techniques: Model compression, quantization, LoRA fine-tuning, adversarial robustness, generative models

Local & On-Device AI: Local agent optimization and deployment, small-parameter model evaluation (2B-10B), privacy-preserving on-device inference pipelines

Framework Development: Multi-agent system architecture, LLM-assisted elicitation frameworks, state-machine design, automated evaluation and judging pipelines

Deployment & DevOps: AWS cloud infrastructure, CI/CD pipelines, model optimization for resource-constrained environments

Research Methods: A/B testing, performance benchmarking, experimental design, statistical validation

Data Analysis: Statistical analysis, predictive modeling, time-series analysis, large-scale data processing

Hardware Integration: Xilinx, mobile chipset optimization (Mediatek, Qualcomm), embedded systems programming

MENTORING & TEACHING EXPERIENCE

Summer Science Discover Mentor | CU Anschutz | July – Aug 2026

- Mentored 2 high school students on Clinical AI research projects on PLACID and cancer severity detection
- Received Letter of Acknowledgement from program organizers in recognition of mentorship contributions to the Summer Science Discover Program

Graduate Student Mentor | *Lenovo Research* | 2019 – 2022

- Mentored 6+ graduate students and summer interns on AI research projects with AI camera applications
- Managed contractor teams and deliverable timelines for clinical AI development projects

Teaching Assistant | *Engineering Design Division, Binghamton University* | 3 Years

- Taught undergraduate courses in signal processing and senior capstone projects
- Developed curriculum materials focusing on engineering basics (eg, coding, CAD programming, circuit design, etc)
- Supervised student projects in engineering technology development
- Consistently rated highly by students in end-of-semester evaluation

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

AI Certifications:

- Deep Learning: Hands-on Artificial Neural Network | SuperDataScience | June 2019
- Data Analyst Certification | IBM | March 2019
- Python for Data Science | February 2019

CITI Certifications: Completed Group 2 Social & Behavioural & Health Information Privacy and Security (HIPS) training

AWARDS & RECOGNITION

- **Letter of Acknowledgement** | CU Anschutz Summer Science Discover Program | 2026 | Recognition for mentorship of high school research students
 - **ACM AI Summit Travel Grant** | CU Anschutz | 2026 | Travel support for the conference attendance
 - **Innovations Fellowship** | CU Anschutz | FY26/27 | CU Innovations fellowship
 - **SVP Individual Excellence Award** | Lenovo | FY24/25 | "Outstanding contributions to AI-camera automation and enterprise feature development"
 - **Teaching Excellence Award** | Binghamton University | 2022
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PROFESSIONAL MEMBERSHIPS & SERVICE

- Institute of Electrical and Electronics Engineers (IEEE) Membership
 - SIGCHI ACM Membership
 - IMIA-SEP Member
 - Post-Doctoral Associate Outreach Member, CU Anschutz
 - **Review Service:** IEEE Access, IEEE Globecomm, Springer Nature, IEEE AGERS, PLOS Biology, HAI
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